



11869 - SPROUT: Linking Parsec-Scale Physics to the Assembly of the First Galaxies

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST has revealed that even faint, low-mass galaxies at $z>6$ are surprisingly abundant, metal-enriched, and structurally complex—challenging our understanding of star formation, feedback, and black-hole growth at Cosmic Dawn.

To explain these discoveries, SPROUT will develop and apply a new generation of ultra-high-resolution cosmological simulations that resolve, rather than parameterize, the key physical processes governing galaxy formation. At parsec-scale resolution, the simulations will follow individual supernovae, stellar radiation, cosmic rays, and metal enrichment from the first (Population~III) stars to the emergence of galaxies and intermediate-mass black holes.

Dwarf galaxies pose a timely and compelling avenue. Every galaxy began its life as a dwarf, making them the natural “sprouts” of cosmic structure. They are the first sites of star formation and the most computationally accessible laboratories for testing galaxy formation physics today. The project will shine a light on the the physics driving the rapid enrichment seen in JWST galaxies, provide evolutionary pathways from star forming clumps to proto-globular clusters, predict if Little Red Dots can emerge from stellar mass black hole seeds in dense dwarf galaxies and constrain the contribution of dwarf galaxies to the photon budget of Reionization. The detailed simulations outputs will be postprocessed to mock NIRCам imaging and NIR-Spec spectra, enhancing the comparison and legacy value of JWST data for the next decade.

OBSERVING DESCRIPTION